Module 1
Foundations of Tool Calling and Chaining

Welcome to the Course
- Course Introduction
  Video • 2 min
- Course Overview
  Reading • 3 min
- Helpful Tips for Course Completion
  Ungraded Plugin • 1 min
- RAG and Agentic AI Professional Certificate Overview
  Video • 5 min

Introduction to AI Agents
- What are AI Agents?
  Video • 12 min
- Comparing AI System Designs
  Reading • 5 min
- When to (and not to) use AI Agents
  Reading • 5 min
- Practice Quiz: Introduction to AI Agents
  Practice Assignment • Grade: 100%

Getting Started with Tool Calling
- Tool Calling for LLMs
  Video • 4 min
- Why AI Needs Tools: From Guessing to Real-World Action
  Video • 4 min
- Tools, Agents, and Function Calling in LangChain
  Reading • 10 min
- Practice Quiz: Getting Started with Tool Calling
  Practice Assignment • Grade: 100%

Building and Orchestrating Tools
- Build Effective AI Tools for Advanced LLMs
  Video • 8 min
- Build Intelligent Agents for Dynamic LLM Tool Use
  Video • 8 min
- Build a Custom Math Toolkit Agent with LangChain
  Video • 6 min
- Lab: Build an AI Math Assistant with LangChain Tool Calling
  Ungraded App Item • 45 min
- Popular Built-in Tools in LangChain
  Reading • 5 min
- Practice Quiz: Building and Orchestrating Tools
  Practice Assignment • Grade: 100%

Module Summary and Evaluation
- **Summary and Highlights: Foundations of Tool Calling and Chaining**
  Reading • 2 min
- Cheat Sheet: Foundations of Function Calling and Chaining
  Reading • 20 min
- Graded Quiz: Foundations of Tool Calling and Chaining
  Graded Assignment • 21 min
- Reflecting on AI Agents: From Concepts to Capabilities
  Dialogue • 15 min

Module 2
LCEL and Manual Tool Calling in LangChain

Module 3
Using Built-in Agents in LangChain

Summary and Highlights: Foundations of Tool Calling and Chaining

3 XP • Foundations of Responsible AI Strategy

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Congratulations! You have completed this lesson. At this point in the course, you know:

- Use simple LLM features for basic tasks, use workflows for predictable and efficient operations, and deploy agents only when complex reasoning or adaptability is needed.
- AI agents sit at the highest end of the AI complexity spectrum, excelling at tasks that require autonomous decision-making, adaptation, and strategy.
- Use the four-step decision framework—task ambiguity, step flexibility, tool variety, and failure impact—to decide if an AI agent is the right fit for the task.
- Avoid using AI agents for simple, repeatable, or high-risk tasks where errors are costly or predictable; tools can perform better.
- Today's AI agents struggle with reliability, high compute costs, and often need human oversight to avoid hallucinations or missteps.
- Manage risks associated with AI agents by setting boundaries, using logs, monitoring outcomes, and keeping a human in the loop for oversight.
- Effective AI agent architecture includes modular components like memory, tool use, planning strategies, and clear reasoning paths.Tool calling enhances LLM capabilities by connecting them to real-time external data and functionality.
- Embedded tool calling improves LLM accuracy and reduces hallucinations by centralizing tool handling within a dedicated library or framework, replacing error-prone client-side implementations.
- Tools help LLMs access external data and support RAG, enabling the use of the organization's or other specialized databases.
- Tools help process images, audio, and video to enable vision, voice, and multimodal reasoning, manage long conversations, and connect to APIs to perform real-world actions.
- The Zero-Shot ReAct Agent uses zero-shot reasoning to solve tasks it hasn't seen before and works best for simple or well-structured problems.

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